

The Electrical and Computer Engineering Masters Programs at USC is one of the largest in the country. As technology advances at an unprecedented rate, our graduates enter the world with all the skills necessary to land influential positions all over the world.

Our distance learning program, [DEN@Viterbi](#), is an elite online graduate program that ***ranked number one in the country***. Students in this program are not required to be on campus to earn a degree, even though the coursework is identical to that of on-campus students and features the same faculty.

Below you will find the wide array of programs the Electrical and Computer Engineering Department has to offer our Master's students. Click on the following icons for more information.

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Program details
and EE
flowcharts

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**Machine
Learning &
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Program details
and example
schedules

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**Analog,
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**Hardware
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Machine
Learning**

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Thesis Option

The MS Thesis Option is also available for students in the MSEE program. Typically MS Thesis students have a high GPA during their course of study (above 3.50) and have identified a faculty with whom they would like to take undertake a two-semester research project and publish it. In the second or third semester of attendance students will form an MS Thesis Committee using the committee forms on the Graduate School's website.

Requirements:

- Minimum requirement of 28 units
- Four (4) of these units are to be thesis, EE 594A (2) and EE 594B (2)
- At least 16 units, not including thesis, must be at the 500 level or higher, and at least 20 units must be in the major department.
- A maximum of 4 units of 590 Research may also be counted toward the degree in addition to EE 594A

Second Master's Degree

A graduate student who already holds a master's degree from USC may apply up to 4 units toward a second master's degree with the permission of the chair of the major department. All credit, including the transferred units, must be earned within seven calendar years.

For students who earned their first master's degree at another institution, no coursework may be repeated from the first program of study and no unit credit from the first program of study may be counted toward the second master's degree.



Students interested in achieving their Masters of Science degree online may do so by enrolling in our top ranked Distance Education Network (DEN) program. For more information regarding our DEN@Viterbi program, please click [here](#).

Change of Major Process to the MSEE-General Degree Program (within Viterbi)

This change of major process applies to current Viterbi graduate students interested in changing their major to the MSEE general degree program in the Ming Hsieh Department of Electrical and Computer Engineering.

Students who are interested would need to first meet with an ECE academic advisor at EEB, room 102 to discuss the process of changing major, and to review the MSEE degree program requirements, the curriculum, and course registration.

Requirements to Transfer:

1. Currently enrolled as a Viterbi School of Engineering Graduate student. This process does not apply to non-Viterbi students.
2. Complete at least two EE Fundamental graduate-level classes based on an area of interest or concentration within EE.

Example #1: Students interested in the Controls area must complete EE-482 and EE-510 or EE-503.

Example #2: Students interested in the VLSI track must complete EE-457 or EE-479 and EE-477.

For more information about specific areas and fundamental courses within the MSEE, please visit:

<https://minghsiehece.usc.edu/academics/ms/master-of-science-electrical-engineering-program-details/>

1. Complete these two fundamental classes during the first year as a graduate student at USC.
2. Earn a B or higher in both courses with a GPA of 3.00 or above (EE classes computed only). A grade below a B would not meet the transfer requirement.
3. If the student has met the requirements, he/she must meet an ECE academic advisor again to complete a Change of Major form after final grades have been posted officially in the University system/OASIS or STARS report.

Change of Major Process to the MS in Financial Engineering (MS-FINE) Degree Program (within Viterbi)

This change of major process applies to current Viterbi graduate students interested in changing their major to the MS in Financial Engineering degree program in the Ming Hsieh Department of Electrical and Computer Engineering.



Students who are interested would need to first meet with an ECE academic advisor at EEB, room 102 to discuss the process of changing major, and to review the MS-FINE degree program requirements, the curriculum, and course registration.

Requirements to Transfer:

1. Currently enrolled as a Viterbi School of Engineering Graduate student. This process does not apply to non-Viterbi students.
2. Complete the two EE Fundamental graduate-level classes in MS-FINE, which are EE-503 and EE-518.
3. Earn a B+ or higher in both courses with a minimum GPA of 3.30 or above (EE classes computed in this GPA only).
4. A grade below a B+ would not meet the transfer requirement for this MS-FINE degree program.
5. If the student has met the requirements, he/she must meet an ECE academic advisor again to complete a Change of Major form after final grades have been posted officially in the University system/OASIS or STARS report.

Change of Major Process to the MS in Electrical and Computer Engineering (Machine Learning and Data Science) Degree Program (within Viterbi)

This change of major process applies to current Viterbi graduate students interested in changing their major to the MS in Electrical and Computer Engineering (Machine Learning and Data Science) degree program in the Ming Hsieh Department of Electrical and Computer Engineering.

Students who are interested would need to first meet with an ECE academic advisor to discuss the process of changing major, and to review the MS-ECM degree program requirements, the curriculum, and course registration.

Requirements to Transfer:

1. Currently enrolled as a Viterbi School of Engineering Graduate student. This process does not apply to non-Viterbi students.
2. Students are required to have some multivariate calculus in their background.
3. Complete at least two EE Fundamental graduate-level classes in MS-ECM which are EE-483, EE-503, and/or EE-510.
4. Earn a minimum GPA of 3.50 or above in two EE fundamental courses and have a minimum GPA of 3.50.
5. Must meet all requirements of the new degree (no substitutions).
6. If the student has met the requirements, he/she must meet an ECE academic advisor again to complete a Change of Major form after final grades have been posted officially in the University system/OASIS or STARS report.

These requirements are effective for academic year 2021-2022, and are subject to change in subsequent semesters. Changes from one semester to the next, if any, will typically be small or moderate. Check with a department advisor to verify current requirements if you have any uncertainty.

Change of Major Process to the MS in Quantum Information Science (MSQIS) Degree Program (within Viterbi)



This change of major process applies to current Viterbi graduate students interested in changing their major to the MS in Quantum Information Science degree program in the Ming Hsieh Department of Electrical and Computer Engineering.

Students who are interested would need to first discuss with an ECE academic advisor to understand the process of changing majors, and to review the MSQIS degree program requirements, the curriculum, and course registration.

Requirements to Transfer:

1. Currently enrolled as a Viterbi School of Engineering Graduate student. This process does not apply to non-Viterbi students.
2. Complete at least one of the EE Fundamental graduate-level classes in MSQIS, which are EE-520 and/or EE-514.
3. Earn a B+ or higher in one or both courses with a minimum GPA of 3.30 or above (EE classes computed in this GPA only).
4. A grade below a B+ would not meet the transfer requirement for this MSQIS degree program.
5. If the student has met the requirements, the student must then contact an ECE academic advisor again to complete a Change of Major form after final grades have been posted officially in the University system/OASIS or STARS report.

Prospective Student Admission

Apply to the MS program

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